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Simulation of international space station microgravity directional solidification experiments on columnar-to-equiaxed transition

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Abstract

It is well known that gravity affects solidification of alloys due to the convective effects it induces. As a result, different outcomes are expected if solidification experiments are carried out in near-zero gravity conditions achievable in space. Directional solidification experiments were conducted on board the Material Science Lab (MSL) in the International Space Station (ISS). The experiments, on Al–7 wt.% Si alloys, were carried out with a low gradient furnace (LGF). The LGF is a Bridgman-type furnace insert for the MSL. Numerical simulations for two such microgravity directional solidification experiments are presented and compared with experimental results. A front tracking algorithm to follow the growing columnar dendritic front, and a volume averaging model to simulate equiaxed solidification, were employed simultaneously in a common thermal simulation framework. The thermal boundary conditions for the simulation domain were computed via the temperature readings which were recorded during the experiments. The simulation results include the prediction of columnar-to-equiaxed transition (CET) and average as-cast equiaxed grain diameters, and agreed with the experimental results reasonably. The simulations predict that although an undercooled zone forms ahead of the growing columnar front, thermal conditions in the diffusion-controlled experiments were inadequate to trigger an entirely equiaxed zone without grain refiners.
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Keywords: CET; Directional solidification; Equiaxed dendritic growth

1. Introduction

Solidification of alloy melts can lead to various grain structures, and two different types of dendritic structure are common in typical castings. The dendrites with an elongated shape and large aspect ratio are termed columnar dendrites, and they usually originate at the mould walls and grow in a direction parallel with, but opposite to, the direction of heat flow, creating a columnar zone. Dendrites of lower aspect ratio than columnar ones may form an equiaxed zone. These equiaxed dendrites finally have random as-cast crystallographic orientations because of the stochastic nature of their origin and their ability to rotate freely in liquid, at least during the early stages of their growth. A columnar-to-equiaxed transition (CET) is a feature at the shared boundaries at which both zones finally meet. Often in a casting, equiaxed dendrites nucleate within the bulk liquid ahead of the advancing columnar front. During solidification, if nucleation and growth yield sufficient volume fraction of equiaxed dendrites ahead of the columnar front to block its progress, a CET will appear in the macrostructure of the casting [1]. CET is an important phenomenon which strongly affects the mechanical properties of cast components. For example, directionally solidified turbine blades are manufactured to promote a columnar zone (thus avoiding the CET), because a columnar structure along the length of the turbine blade improves the creep resistance at

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high service temperatures. For most ambient temperature applications, small equiaxed grains enable improved liquid feeding during casting solidification and result in higher yield strength in components.

Solidification of metals and formation of dendritic grains during the casting process involve many physical phenomena. The gravity-related natural convection and buoyancy that are present in solidifying liquid melts can extensively modify important physical mechanisms such as heat, mass and species transport. Over the last few decades, CET has been examined by many experimental and theoretical modeling studies [2]. However, the effects of gravity on natural convection and buoyancy of free equiaxed dendrites add complexity to solidification experiments under terrestrial conditions. So far a limited number of solidification and solidification related experiments [3–12] have been conducted under microgravity conditions and only a few of these have been dedicated to study CET in metallic alloys [9–12].

To exploit microgravity conditions to study CET formation in alloy solidification, the European Space Agency has coordinated a specific Microgravity Applications Promotion (MAP) project called CETSOL (CET in solidification processing). Experiments on board sounding rockets have been used to study CET formation in Al–Si alloys [10–12]. In particular, experiments on a MAXUS (EADS-Astrium, Germany) sounding rocket enabled 10 min of microgravity solidification of an Al–7 wt.% Si alloy. A power-down method, in which moving parts were not required, was used to trigger CET in the alloy [12]. Most recently, the Material Science Lab (MSL) on board the International Space Station (ISS) has been used as a long-duration microgravity platform for CETSOL experiments. The ISS enables multiple experiments by uploading feed samples and downloading as-solidified samples, with the experiments being set up by astronauts and controlled from Earth. To date, upload and download has been via the NASA space shuttle but following the retirement of the shuttle fleet, future transport of CETSOL samples may be via private operators, e.g. SpaceX's Dragon spacecraft. Numerical model predictions of such directional solidification experiments conducted on board ISS-MSL, and comparison and analysis with the experimental results, are presented in this paper.

2. Experimental configuration

The MSL on board the ISS Destiny module consists primarily of a process chamber accommodating furnace inserts together with the individual experiment cartridge and other auxiliary instruments to conduct solidification experiments. Various furnace inserts with experiment cartridges allow for different types of materials science experiments to be performed. The low gradient furnace (LGF) is such a furnace insert to the MSL and is a Bridgman-type furnace that can directionally move relative to the sample cartridge assembly (SCA), which contains the 8 mm diameter, 245 mm long experimental sample. To establish a temperature gradient, the LGF heating chamber has seven independently controlled zones with separate heating elements. An adiabatic zone is placed between the third and fourth heating zones (see Fig. 1a).

The experimental sample is contained in the SCA (see Fig. 1b), which consists of a cartridge head (seen on the left-hand side of Fig. 1b) and a sealed cylindrical sample compartment (cartridge tube) to accommodate the cylindrical sample (Fig. 1b). The SCA head contains mainly the electrical connections but no software and no complex electronics are included. The SCA sample compartment is a sealed tantalum cylinder (the cartridge tube), primarily containing the experiment-specific sample within an alumina crucible. Furthermore, 12 thermocouples (TCs) are attached to the crucible, fixed at 10 mm intervals (dark dots in Fig. 1a) at the mid-section of the sample. The position of the first thermocouple is 72 mm from the non-molten (left hand) end of the metallic sample. These type-N thermocouples were of 0.5 mm outer diameter, and numbered from TC1 to TC12. Thermocouple numbering starts from the cold side, for example TC1 was the left-most thermocouple according to Fig. 1 and TC12 was the right-most one. The compartment outer body is closed by a welded hot end cap on one side, with a stainless steel support. The schematic setup of the SCA in the LGF with relevant (selected) dimensions is given in Fig. 1a. Six experiments have already been conducted on board the ISS, one more will be carried out as a terrestrial experiment and a few more samples were uploaded to the ISS during the recent and final space shuttle mission STS-135. Two of the experiments (named here as FM#2 and FM#3) are considered for the work involved in this paper. FM indicates flight module. The experiments on board ISS were performed during sleeping phases of the astronauts to minimize disturbances of the microgravity condition. Measurements indicated that in all cases a g-level less than 0.001 g was realized.

During the experiment, initially the furnace was moved from the right to the left (in the opposite direction to the arrows in Fig. 1a) to partially melt the sample. The furnace was retained at the left-most position for a reasonably long time period, allowing the temperatures to reach steady state. Thereafter the furnace was moved from left to right, at different given speeds. Temperature readings from the thermocouples attached to the sample crucible (thermocouple number 1 is the left-most one) were electronically recorded during the experiments and are presented in Fig. 2.

The experiments investigated here (FM#2 and FM#3) were conducted with similar experimental conditions, on samples of diameter 8 mm. The temperature readings collected from seven (out of 12) representative thermocouples attached to the crucible are presented in Fig. 2a and b. During both experiments, the furnace was moved initially at the speed (V) of 0.01 mm s^{-1} for 2000 s and then at a speed of 0.2 mm s^{-1} for 250 s, creating two pulling velocity regimes. The main difference between the two experiments is the presence of grain refiners. One sample (FM#2) con-

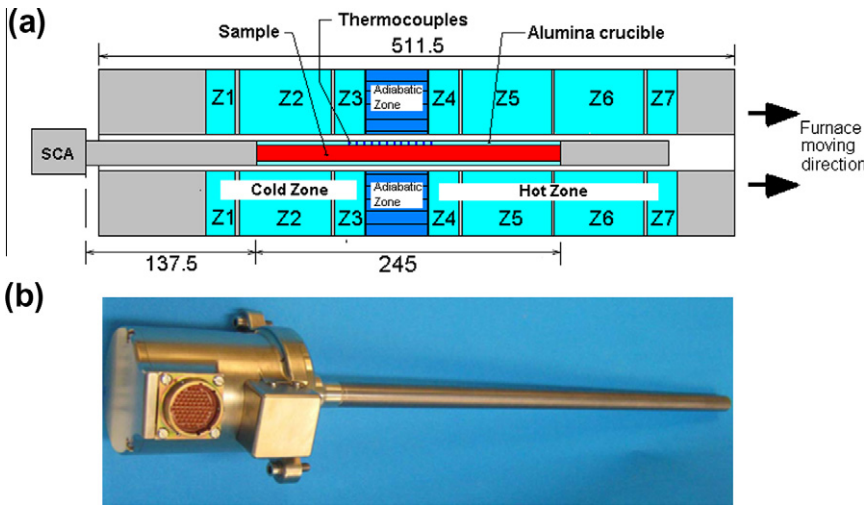

Fig. 1. (a) The schematic setup of the LGF and SCA; all dimensions in mm. (b) Photograph of sample cartridge assembly (SCA) for MSL-LGF.

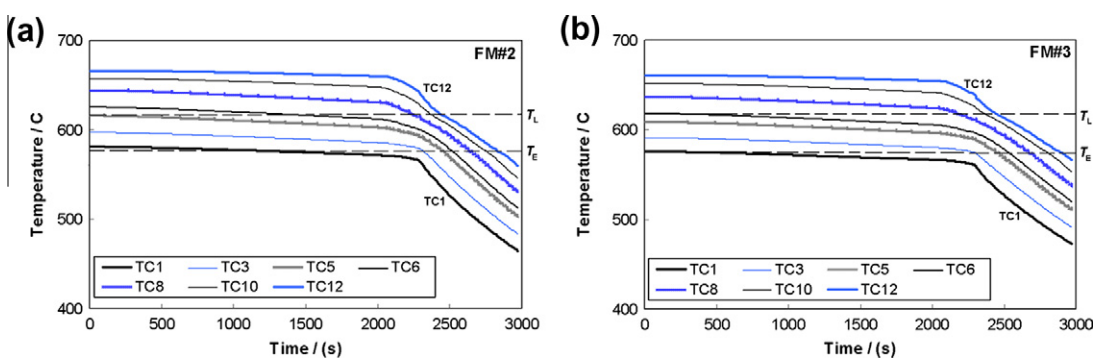

Fig. 2. Thermocouple readings during the experiments for (a) FM#2, and (b) FM#3. The liquidus (T_L) and eutectic (T_E) temperatures of the alloy are also indicated.

tained TiB_2 grain refiners whereas the other (FM#3) was free of grain refiners. Table 1 presents the chemical composition of the experimental samples, nominally Al–7 wt.% Si, from which a grain refiner density can be approximated, as explained in the next section.

3. Computational model and input parameters

The two microgravity experiments presented in Section 2 were numerically simulated to predict CET and analyze solidification conditions. The front tracking (FT) approach originally developed by Browne and Hunt [13] was employed to model the columnar growth – exploiting the model's ability to track the advance of the columnar dendritic front together with evolution of the undercooled zone ahead of it. The algorithm uses massless markers to represent the curve joining the tips of the columnar dendrites [13,14], which are advanced according to the kinetics of dendrite growth. This has been shown to be a very good approximation for growth of aligned dendrites [15]. An equiaxed solidification model [16] was also integrated to the solidification model and the combined model can simulate the equiaxed solidification ahead of the advancing columnar front. The model is thus capable of predicting the CET. This combined modeling approach has already demonstrated the ability to integrate effects of gravity [17].

The initial condition for the solidification model is an alloy which is partially melted due to the steady state thermal gradient. Solidification starts when the furnace (see Fig. 1a) starts to move relative to the SCA. Epitaxial type growth initiation from the existing solid is assumed for the columnar solidification in the experiments. Therefore the columnar markers for the FT simulation were distributed along the liquidus isotherm at the very first time step. This marker (columnar front growth) velocity is calculated using the dendrite tip velocity, v_t–undercooling, ΔT relationship as follows [18]:

$$v_t = C(\Delta T)^n \tag{1}$$

where C and n are alloy-dependent constants and for Al–7 wt.% Si, $n = 2.7$ and $C = 2.9 \times 10^{-6}\ \mathrm{m\,s^{-1}\,°C^{-n}}$ [18]. The exact temperature at each marker particle was computed by interpolating local control volume temperatures according to the position of the marker particle.

The FM#2 sample contained TiB_2 grain refiners. It has been shown that TiB_2 grain refiner particles in typical

Table 1
Chemical composition of the alloy samples.

Sample	Percentage	ppm																					Al
	Si	Ni	Fe	Cu	Mn	Mg	Cr	Zn	Ti	B	Be	Na	Ca	Li	Zr	Bi	Ag	Ga	Sn	P	Pa		
FM#2	7.04	2.1	11	15	<0.5	1.4	<0.5	7.5	215	30	<0.2	<0.2	<0.5	<0.2	<0.5	<2	15	<0.5	<3	<5	<1	Rest
FM#3	6.99	<0.5	2	7.1	<0.5	1.6	0.3	8.1	5.5	0.8	<0.2	<0.2	<0.5	<0.2	<0.5	<3	4.8	<0.5	<3	<5	<1	Rest

aluminum alloys follow a log-normal size distribution [19]. The particle size distribution can be presented mathematically as follows, where particle diameter is given by d:

$$f(d) = \frac{1}{\sigma d\sqrt{2\pi}} \exp\left(-\frac{(\ln d - \ln \varphi)^2}{2\sigma^2}\right) \tag{2}$$

where the standard deviation and the geometric mean of the particle size distribution are denoted by σ and φ, respectively.

To start free growth from a particle of a diameter d, the critical undercooling ΔT_{cr} required is [19]

$$\Delta T_{cr} = \frac{4\gamma}{\Delta S_v d} \tag{3}$$

where ΔS_V is the volumetric entropy of fusion and γ is the solid–liquid interfacial energy. The total number of equiaxed nucleation events triggered by a given point in time is computed by considering the current undercooling, the cumulative distribution function for Eq. (2) and the average grain refiner density [16]. Considering a circular equiaxed envelope that wraps around the primary dendrite tips for the two-dimensional (2-D) simulation, growth and impingement of the equiaxed dendrites were computed (as detailed in Ref. [16]). For dendrite tip growth rate computations, both columnar and equiaxed, the relationship given in Eq. (1) was applied. When a coherent equiaxed network is formed ahead of the columnar front, a physical barrier to the advancing columnar front forms and blocks further advancement [17]. However, if the equiaxed dendrite diameters are very small (less than the primary columnar dendrite arm spacing, computed according to Ref. [20]) the columnar dendritic front is expected to progress by encapsulating such small non-coherent equiaxed dendrites between primary columnar dendrite arms.

As gravity is negligible on the ISS, diffusion of heat dominates solidification, at grain level, of both dendrite types [21]. The energy equation with the latent heat source term was used as follows:

$$\rho C_p \frac{\partial T}{\partial t} = \nabla(K\nabla T) + L\frac{\partial f_s}{\partial t} \tag{4}$$

where ρ is density, K is thermal conductivity, C_p is specific heat, f_s is solid volume fraction, and T and t denote the temperature and time, respectively. The variations of K and C_p with T were computed according to the polynomials in T detailed in Refs. [14,18].

For the axial coordinate z, the left-most end of the sample was taken as the datum, i.e. $z = 0$. For the simulations, a 132 mm long section from the experimental sample which starts from $z = 61.5$ mm was considered. To define the thermal boundary conditions for the simulation domain, the thermal data recorded during the experiments on board ISS-MSL were used. These data were collected from the 12 thermocouple positions placed along the alumina crucible, at 10 s intervals; see Fig. 2. The interface heat transfer between the crucible and the sample was computed by

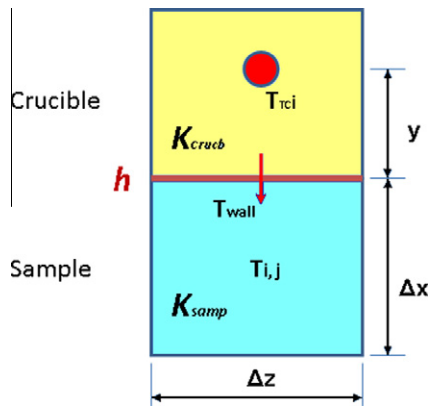

Fig. 3. Control volumes, representative of the axial direction at thermocouple locations, either side of the crucible/sample interface, used to consider interface heat transfer between crucible and sample.

the inverse heat transfer method [14], by considering appropriate experimental parameters (as shown in Fig. 3). Here T_{Tci} and $T_{i,j}$ represent the node temperatures in the crucible and the sample, respectively; h is the interfacial heat transfer coefficient; and Δz, Δx and y are relevant distances. Intermediate spatial and temporal input thermal data values used in the simulations were linearly interpolated/extrapolated from the available experimental data.

To calculate average grain refiner particle density in the FM#2 sample, it was assumed that all boron atoms are contained in TiB_2 particles. This assumption converts TiB_2 concentration to 15 ppm. According to the previous experimental data [19,22], 1 ppt TiB_2 in Al–7 wt.% Si provides 8.6×10^{12} particles m^{-3} of the Al alloy. The value of 15 ppm is therefore equivalent to 129 particles mm^{-3}. Considering the maximum 1% efficiency of TiB_2 grain refiners [19,22], the 129 particles mm^{-3} figure was normalized to the effective value of 1.25 (efficiency ~0.97%) particles mm^{-3}. Unit depth relevant to the 2-D modeling domain of 8×132 mm will therefore contain ~1320 effective grain refiner particles. But model computations presented here are 2-D and so we need to consider that only grains nucleated within a thin slice (approximately equal to the final average grain radius) can contribute to our simulated microstructure appearing in two dimensions. Therefore, it was assumed that a slice with thickness equal to half of the average equiaxed dendrite diameter contributes to the 2-D view; the as-cast average equiaxed dendrite diameter is ~1.5 mm. Thus, the input grain refiner density for the 2-D model is obtained by multiplying the three-dimensional (3-D) grain refiner density value by a factor equal to 0.75 mm^{-1} (here 0.75 mm is approximately one half of the average grain diameter). Thus, a total number of 990 equally spread grain refiner particles was used as a simulation parameter for FM#2. Typical particle size distributions of TiB_2 and other thermo-physical data related to TiB_2 particles that we used as simulation parameters were obtained from Ref. [19].

The respective set of thermocouple readings collected during the experiments (Fig. 2) was used as the input data to define the boundary conditions (Fig. 3) for each simulation. For all of the simulations, alloy related thermo-physical properties were taken from Ref. [18]. Further simulations were conducted with the columnar FT model alone to analyze the conditions ahead of the columnar front for the two experiments considered here.

4. Experimental results

A NASA space shuttle mission (STS-130) downloaded the SCAs back to Earth after the experiments. The samples were cut into several sections for metallographic analysis of the microstructure, each of length ~30 mm. Initially the microstructure of radial cross-sections was revealed. Then the sample sections were in turn cut into two halves and the microstructure of the axial cross-sections were also revealed. Fig. 4 shows a sketch of FM#2 and FM#3 flight samples with related experiment phases and identification of the axial (longitudinal) cross-sections L1 to L6 for the metallographic preparation. Solidification phase 1 is at low puling speed, phase 2 at the higher speed, and the final fast movement quenches the sample. To analyze the grain structure the axial cross-sections were electrolytically etched. Observations using a polarizing microscope show grains with different crystallographic orientations in different colors. Thus the grain structure and the size of the grains can be determined.

The microstructure of axial cross-sections L2 to L6 for FM#2 and FM#3 are presented in Fig. 5a and b, respectively. The boundary between each section along the sample was nearly planar and transverse to the central axis. Fig. 5a(ii) and (ii) shows the sample sections that contain the end of the temperature gradient zone melted (TGZM) zone (left parts of each axial section) and the beginning of the solidification zone for the FM#2 and FM#3, respectively.

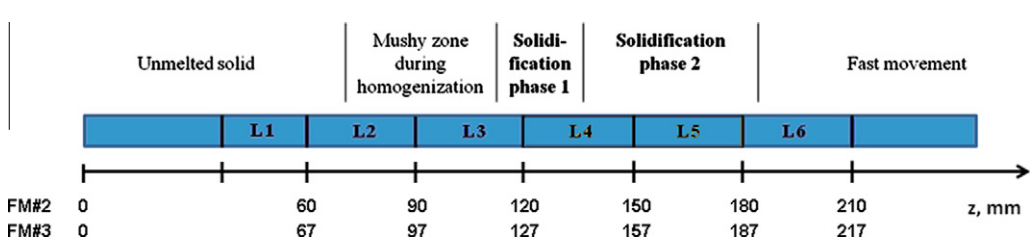

Fig. 4. Sketch of FM#2 flight sample with related experiment phases and identification of the axial cross-sections for the metallographic preparation.

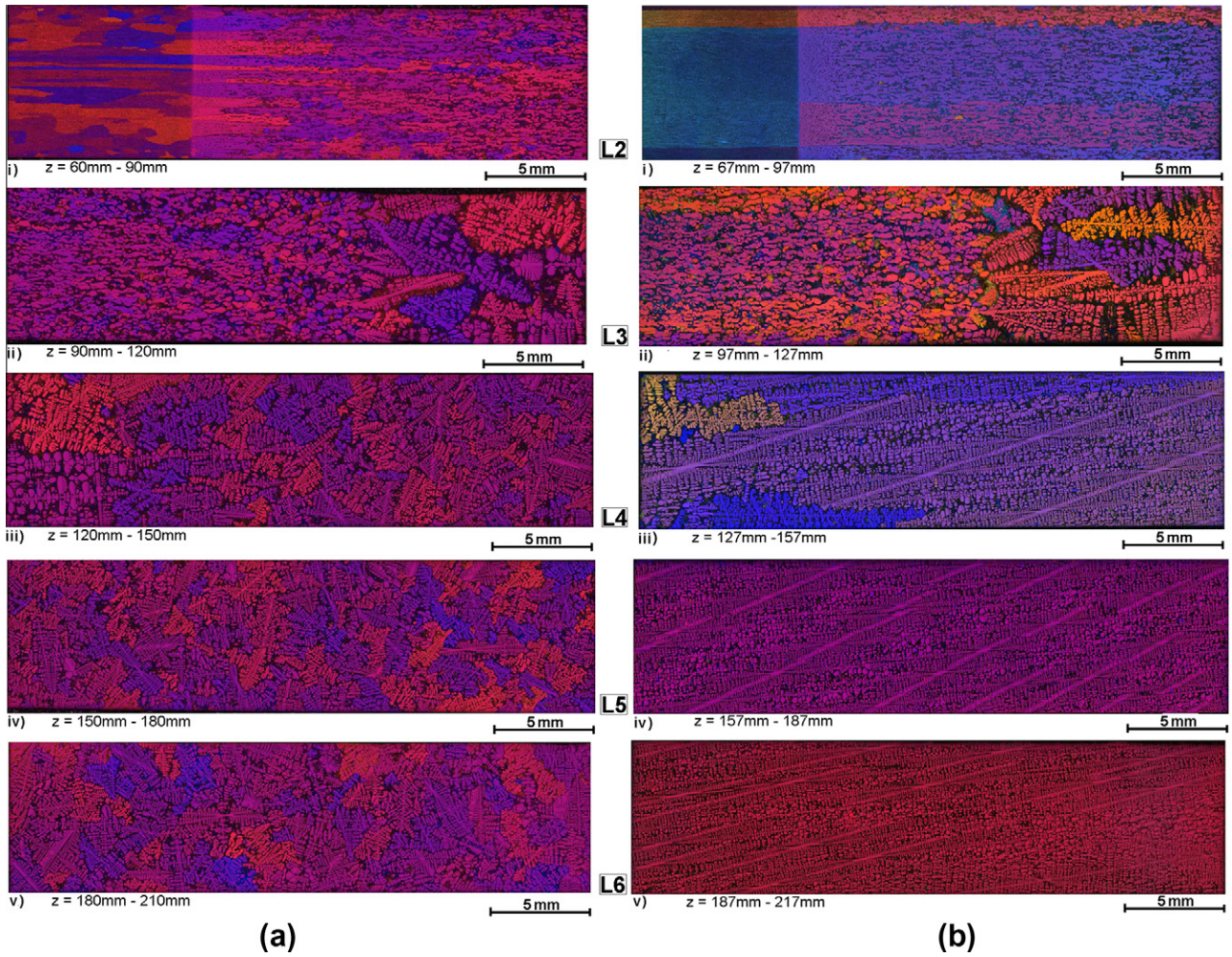

Fig. 5. Microstructure: axial cross-sections of (a) FM#2 and (b) FM#3. Column 1 shows the FM#1 microstructure; column 2 is of FM#3. The rows (i) to (v) are the sections of L2 to L6.

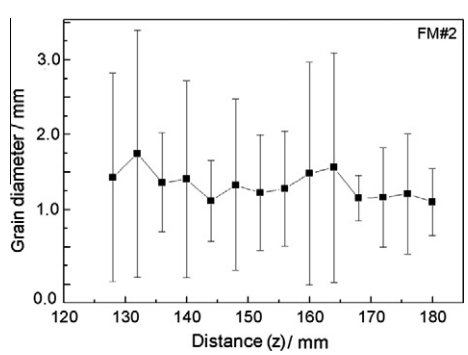

Fig. 6. Experimental equiaxed grain diameters for FM#2.

The TGZM zone, which was formed from a semi-solid region, developed through melting and the temperature holding period prior to solidification. The temperature holding period for both of these experiments was ~300 min. The boundary between the two zones is at a distance of approximately $z = 110 \pm 2$ mm for FM#2 and for FM#3 it is 116 ± 2 mm. In both experiments solidification was initiated in a columnar form. For FM#2, CET is found to be at a distance of $z = 126 \pm 2$ mm (see Fig. 4a(iii)). A detailed analysis of determination of this CET position is given in Ref. [23]. From the CET position onwards the rest of the sample is fully equiaxed with no significant variation of equiaxed grain size along the axial direction. For FM#3 no transition to equiaxed growth is observed, but it seems that a single dendritic crystal has emerged by competitive growth from the columnar zone.

The mean area of the equiaxed grains is ~1.5 mm^2 for the FM#2 experiment. Fig. 6 shows the measured equiaxed grain diameter variation along the axial direction of the sample based on ~15–30 single grains for each position. As a result the mean grain size is within the range 1.1–1.8 mm. Therefore the diameter was taken as 1.5 mm for the 2-D to 3-D conversion factor outlined in Section 3.

5. Simulation results

5.1. CET prediction

The combined columnar and equiaxed solidification model was used to simulate the FM#2 experiment. The

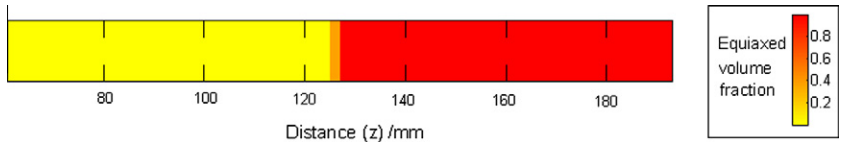

Fig. 7. Predicted final as-cast equiaxed volume fractions and therefore CET for FM#2.

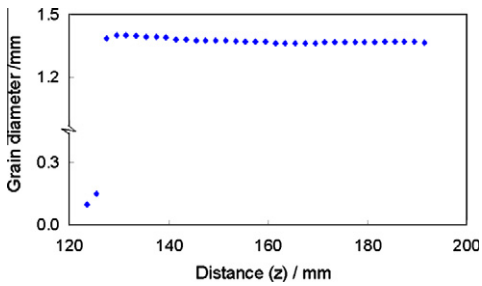

Fig. 8. Predicted average equiaxed grain diameter along the axial direction for FM#2.

temperature data from this experiment and grain refiner size distribution parameters from Refs. [19,22] were used as inputs. Fig. 7 presents the predicted as-cast columnar and equiaxed volume fractions from the simulation. The predicted equiaxed grains start to appear at $z = 124.5$ mm in the simulation and at $z = 127.5$ a fully equiaxed region begins, giving the CET. Equiaxed grain diameters are also predicted from the simulation and the predicted variation of average grain diameter along the axial direction is shown in Fig. 8. Even though some equiaxed grains appear in the zone between $z = 124.5$ mm and $z = 127.5$ mm giving small equiaxed grain diameters, the volume fraction of equiaxed grains is considerably low in this region; they are embedded between columnar grains.

5.2. Simulation of columnar front progress

The initial (at the first time step) liquid/solid interface positions are at $z = 109$ mm and $z = 116$ mm for FM#2 and FM#3 simulations, respectively. Fig. 9 presents the columnar front positions and liquidus isotherm positions obtained from the columnar FT simulations (with equiaxed solidification purposely disabled in the model) conducted using the temperature data sets of FM#2 and FM#3 experiments. Fig. 10 presents the average thermal gradient (G) at the columnar front, obtained from the same simulations. Here thermal gradient was obtained by considering the simulated temperature field (extended to 6 mm) ahead of the columnar front.

6. Discussion

During the experimental sequence initial melting of the sample was conducted by moving the Bridgman furnace towards the sample (to the left; Fig. 1a). The superheat generated from the hot zone of the furnace melted the sample back to certain distance. The temperature holding period, ~300 min for each experiment, can be assumed as a reasonable time to have established quite a stable TGZM region. To simulate the exact position of the solid/liquid position at the beginning of the solidification part of the experiment, an input from an alloy melting model would be ideal. Unfortunately, no advanced alloy melting model exists to date to the best of our knowledge (e.g. Ref. [24]), and indeed the use of the enthalpy method has been deemed sufficient to simulate the melting phase of welding [25]. The primary concern of this contribution is modeling and analysis of the CET in solidification and no particular attention was paid to modeling the melting process. This is because, unlike in welding, here the melt-back is stabilized as a steady state prior to the triggering of the solidification experiment. After a long stabilization period it is rational to assume that no solid exists in the regions with temperatures above the liquidus temperature. Thus, initial dendrite tip/liquid interface remains at the liquidus isotherm of the sample. Indeed, this is also consistent with Eq. (1) where, for zero front velocity the undercooling is zero and vice versa. Comparing the interface position at the initial time step of the simulations (Section 5.2) and the experimental results (Section 4) as shown in Fig. 5a(ii) for FM#2 and Fig. 5b(ii) for FM#3, confirm that the experimental and simulated initial conditions are in agreement. So the assumptions on initial conditions are acceptable.

Comparing the experimental results with the simulation for FM#2, the columnar front was assumed to progress from the liquidus isotherm at the start of the simulation. Both front and liquidus are co-located for a period, indicating the lack of a region of significant undercooling in the bulk liquid. The liquidus starts to advance towards the liquid and then, diverging from the columnar front, an undercooled liquid zone develops ahead the front, thus facilitating equiaxed nucleation. In the simulation, equiaxed nucleation from grain refiners was considered according to the free growth model [22]. Grain refiner particles may be engulfed or pushed by the growing solid/liquid interface during solidification of catsings. Local reductions in undercooling due to latent heat and solutal effects can also "poison" grain refiner action, making them ineffective. So, most grain refiner particles cannot initiate grains and some may end up in the grain boundaries [19], yielding general grain refiner efficiency at a very low level. Nevertheless, the efficiency of the grain refiners is at a maximum of 1% when addition level is low [19]. As found from the calculations given in Section 3, in the FM#2 experiment the 15 ppm TiB_2 count is a relatively low grain refiner addition

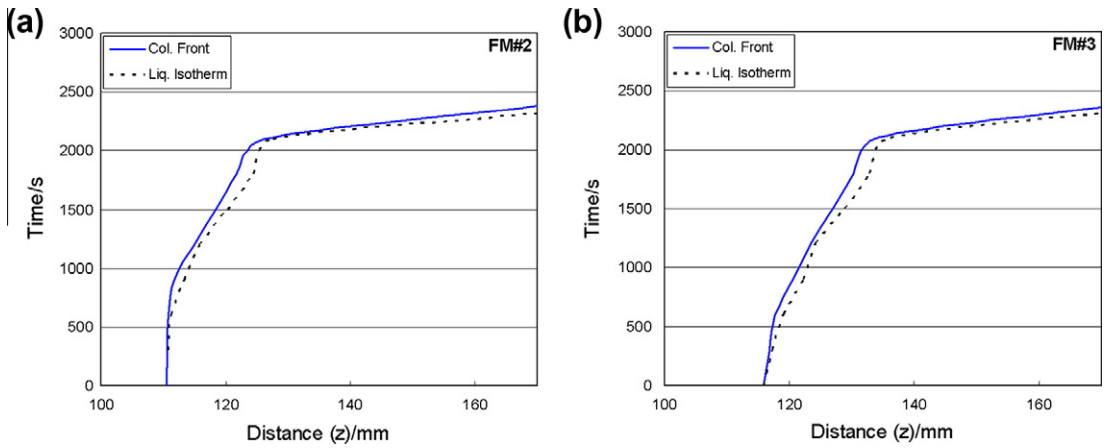

Fig. 9. Predicted columnar front and liquidus isotherm positions: (a) for FM#2 and (b) for FM#3.

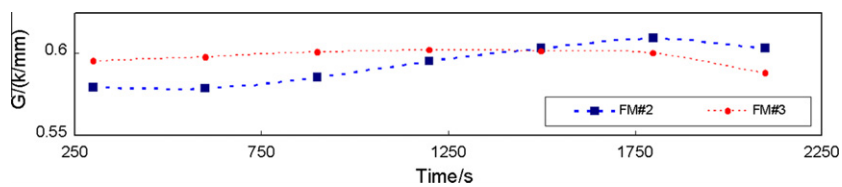

Fig. 10. Average numerically predicted thermal gradient at the columnar front.

quantity. Therefore it was assumed near 1% maximum efficiency of grain refiners. The 3-D to 2-D grain refiner density conversion factor was obtained by considering a slice of a sample that can contribute to the 2-D microstructure observation. Here dendrites that nucleate with a thickness equal to half of the (average) grain diameter were considered to contribute to the 2-D view, giving the factor of 0.75 mm^{-1}.

An analysis of the sensitivity of grain refiner efficiency for the simulation results was also conducted. The CET position was not very sensitive to slight changes of grain refiner efficiency around the 1% value; however, the predicted equiaxed grain size was altered considerably. For example, 2% grain refiner efficiency reduced the average grain diameter in the equiaxed zone to ~0.5 mm and the predicted CET was at $z = 127.1$ mm. Similarly, slightly lower (0.9% and 0.85%) grain refiner efficiencies made the equiaxed grains larger and shifted the CET position marginally towards the equiaxed region ($z = 127.7$ mm and $z = 128.1$ mm respectively). Very low (0.01%) grain refiner efficiency was not sufficient to produce any CET in the simulations, consistent with FM#3 experiment results. Very high grain refiner efficiency values were not considered as they are not realistic for actual alloy solidification [19]. Intermediate values between 0.8% and 0.01% efficiency yielded different CET positions and equiaxed grain diameters, neither of which were close to the experimental figures from FM#2. Therefore, the pre-assumed near 1% efficiency (0.97%) for grain refiners appears to be reasonable: as the value was varied to 0.97% from above and below a convergence of both the predicted CET position and the average equiaxed grain diameter to the experimental values occurred.

During solidification, when the required undercooling – as per Eq. (3) – is available ahead of the columnar front, equiaxed dendrites can start to grow from the grain refiner particles. If the columnar front passes equiaxed dendrites which are small and not numerous enough to block the columnar growth, it can seize such immature equiaxed crystals by encapsulating them between the columnar trunks. Such examples have been cited before in experimental results [26] as well as in simulations [27]. Of course such trapped equiaxed dendrites can grow in an elongated manner together with the columnar grains without leaving much trace of their origin as equiaxed dendrites. Thus any post-mortem microstructure examination may identify those as columnar. Finally, eventual equiaxed domination will block the columnar growth and trigger CET. Following this scenario in the model simulation, CET is predicted at a distance of $z = 127.5$ mm as shown in Fig. 7. Simulation predicts some relatively small (diameter less than 0.3 mm) equiaxed grains before the end of the columnar zone and then grains with diameters of ~1.4 mm throughout the equiaxed zone. The experimental results for the FM#2 sample shows the measured equiaxed diameters in the approximate range between 1.1 mm and 1.8 mm (Fig. 6), giving the average value close to 1.5 mm.

The FM#2 columnar length is ~16 mm in the axial direction and is just slightly less than the distance that the furnace traveled at a low velocity (0.01 mm s^{-1} × 2000 s).

Therefore one can argue that the CET is associated with the pulling velocity jump in the experiments. However, with the same pulling velocity jump, FM#3 did not exhibit a CET. The microstructure remains fully columnar and the dendrite spacing decreases due to the increasing solidification velocity. We have examined the conditions ahead of the growing columnar fronts with the columnar front tracking model as mentioned in Section 3. According to Fig. 9, from the beginning until approximately $t = 500$ s of the solidification sequences for both experiments, the undercooled zone barely existed. The gradual increase of undercooled zone size with some fluctuations can be observed for the duration from approximately $t = 500$ s until $t = 1500$ s. Then from there, the undercooled zone starts to become wider and the maximum width is observed at the solidification time just below and up to $t = 2000$ s. After $t = 2000$ s the undercooled zone shrinks. During this complete solidification time, the computed thermal gradient at the columnar front was within the 0.55 K mm^{-1} to 0.6 K mm^{-1} range for both experiments (Fig. 10). Only a very slight decrease in G at $t = 2000$ s is observed. Therefore it can be seen that the influence of thermal gradient at the columnar front on CET from these experiments is not a dominant one. The wider undercooled zone provides the opportunity to nucleate and grow equiaxed dendrites. The widest undercooled zone appeared at around $t = 2000$ s for both the experiments and in FM#2 it was at the distance, $z = 126 \pm 2$ mm and for FM#3, $z = 132 \pm 2$ mm. Typically, grain refiner particles initiate free growth of equiaxed grains at quite a low undercooling. Equiaxed dendrites that were born early at a low undercooling had relatively higher undercooling dwell times before the columnar front reaches them. Such a relatively extended period allows equiaxed dendrites to establish an equiaxed zone and trigger CET [28]. Even though similar thermal conditions existed ahead of the columnar fronts in both experiments, particles large enough to act as substrates to initiate equiaxed growth were absent in FM#3 melt. As solidification progressed, the undercooled zone became ~3–4 mm wide (see Fig. 9) in both experiments. The approximate columnar length of 16 mm in FM#2, and equiaxed zone formation, appears soon after the pulling velocity change. It implies the pulling velocity increase triggered the CET, but only in the presence of grain refiners. Conditions of relatively high values of the ratio V/G have been shown to favor equiaxed growth [1,29,30]. However, as in experiment FM#3, increasing V alone without grain refiners was not enough for CET to occur. In the absence of convective flow, the effect of solute enrichment cannot exceed the respective solutal diffusion length, l_s (here $l_s = d_l/v_t$, where d_l is the diffusivity of solute in the liquid) [31]. The maximum solutal diffusion length remains at ~75 μm for both the solidification sequences (where $d_l = 6.4 \times 10^{-9}$ m^2 s^{-1}) according to the computed growth rates; this is negligible when compared to the width of the undercooled zone. Under this scenario, the solute that is rejected from the columnar front can not have an effect on the equiaxed nucleation and growth in the undercooled zone even at a modest distance ahead the columnar front and vice versa. Therefore, the solute effects on CET are deemed insignificant in the considered experiments here. It has been shown [21] that thermal interactions are more important than solutal ones in determining grain boundaries when grain size is of the order of millimeters.

In general the evaluation of samples processed in a microgravity environment allows for quantitative investigation of the CET under diffusive conditions. In contrast, physical phenomena such as natural thermo-solutal convection and buoyancy effects will have a considerable influence on the final grain structure if similar experiments are conducted under the terrestrial conditions. Samples processed on Earth with identical parameters presented here are not available yet. A previous set of different experiments [32], performed in microgravity during a sounding rocket mission and also in normal gravity on Earth, has demonstrated that reduced gravity promotes CET. Apart from that, equiaxed grain motion (e.g. Ref. [33]) due to buoyancy effects and convective flow under the terrestrial conditions also contributed to the equiaxed zone formation dynamics and the CET. Melt convection has long been considered as a potent mechanism for dendrite fragmentation [34] while in situ X-ray studies of Al–Cu alloy solidification [35] has shown that gravity-induced flow and buoyant forces are responsible not only for fragmentation but also for the transport of fragments to form an equiaxed zone and appearance of the CET. As with the simulations, the presence of nuclei for heterogeneous equiaxed grain nucleation is a crucial requirement for the CET. Thus, both favorable and unfavorable conditions for CET are present in a ground experiment similar to FM#3. Pulling velocity jump and the presence of grain refiners were critical for the CET in FM#2.

7. Conclusion

Using a combined front tracking/volume averaging model of solidification, numerical simulation of two MSL-LGF microgravity solidification experiments and respective experimental results were presented. Thermal data collected during the experiments were used as input to create thermal boundary conditions for the simulations. Realistic grain refiner data were employed effectively as input to predict CET and equiaxed grain sizes in the experimental sample, FM#2, that contained grain refiners. The simulated prediction of CET position is in very close agreement with that measured experimentally. To compare the solidification conditions in the other experiment, FM#3 simulations for columnar front progress were conducted. It was found that the thermal gradient and formation of undercooled zone ahead of the columnar front were similar for both experiments. The pulling velocity increase alone was not enough to stabilize equiaxed zone in the studied experiments and the effect of grain refiners was crucial for the CET under the given conditions.

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